CS & IT ENGINEERING

Digital Logic

Combinational Circuits

Lecture No. 9

By- Aditya sir

Recap of Previous Lecture

Topic

Decoder

Questions

Topics to be Covered

Topic

Combinational Circuits

→ Adder, Subtractor

Adder → Half, Full

Subtractor → Half, Full

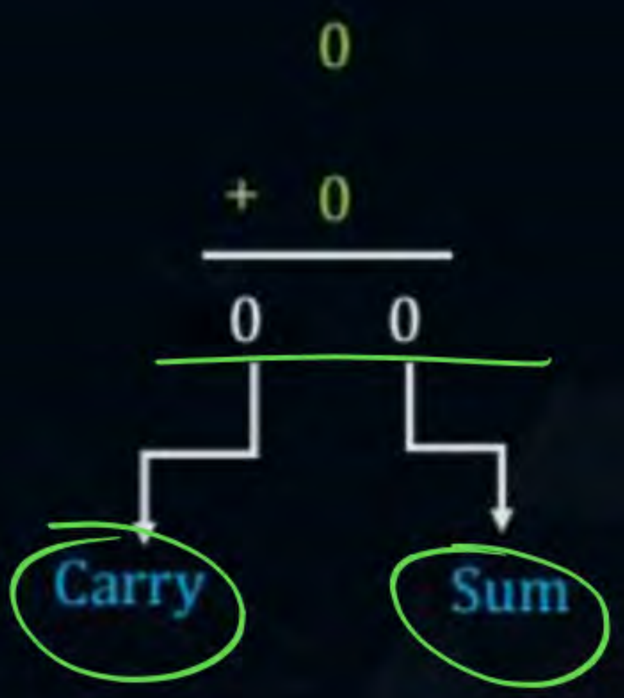
HA / HS $\longrightarrow$ 2 bits i/p

FA / FS $\longrightarrow$ 3 bits i/p

Topic : Half Adder

Half Adder

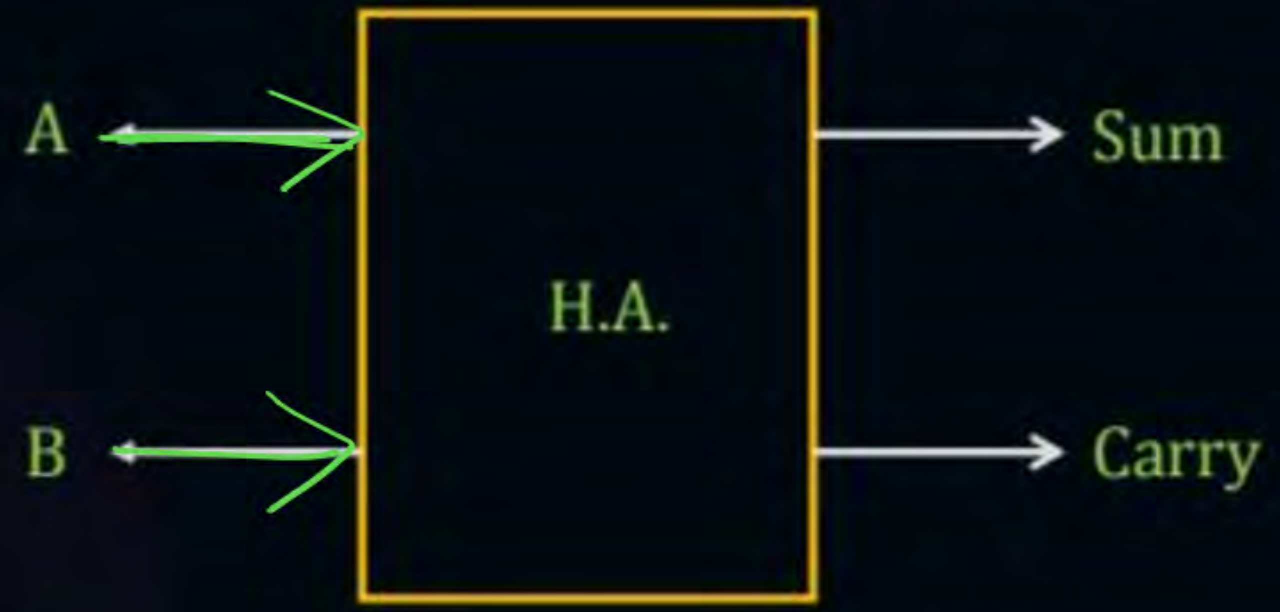
0	0	1	1
+0	+1	0	1
00	01	01	10



- Two bit adder are known as half adder.

Topic : Half Adder

Step-1



2 i/p

2 o/p → 1) Sum
→ 2) Carry

Topic : Half Adder

Step-2

Truth Table

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Topic : Half Adder

Step-3

$$\text{Sum} = \overline{A}B + A\overline{B} = A \oplus B$$

$$\text{Carry} = AB$$

Step-4

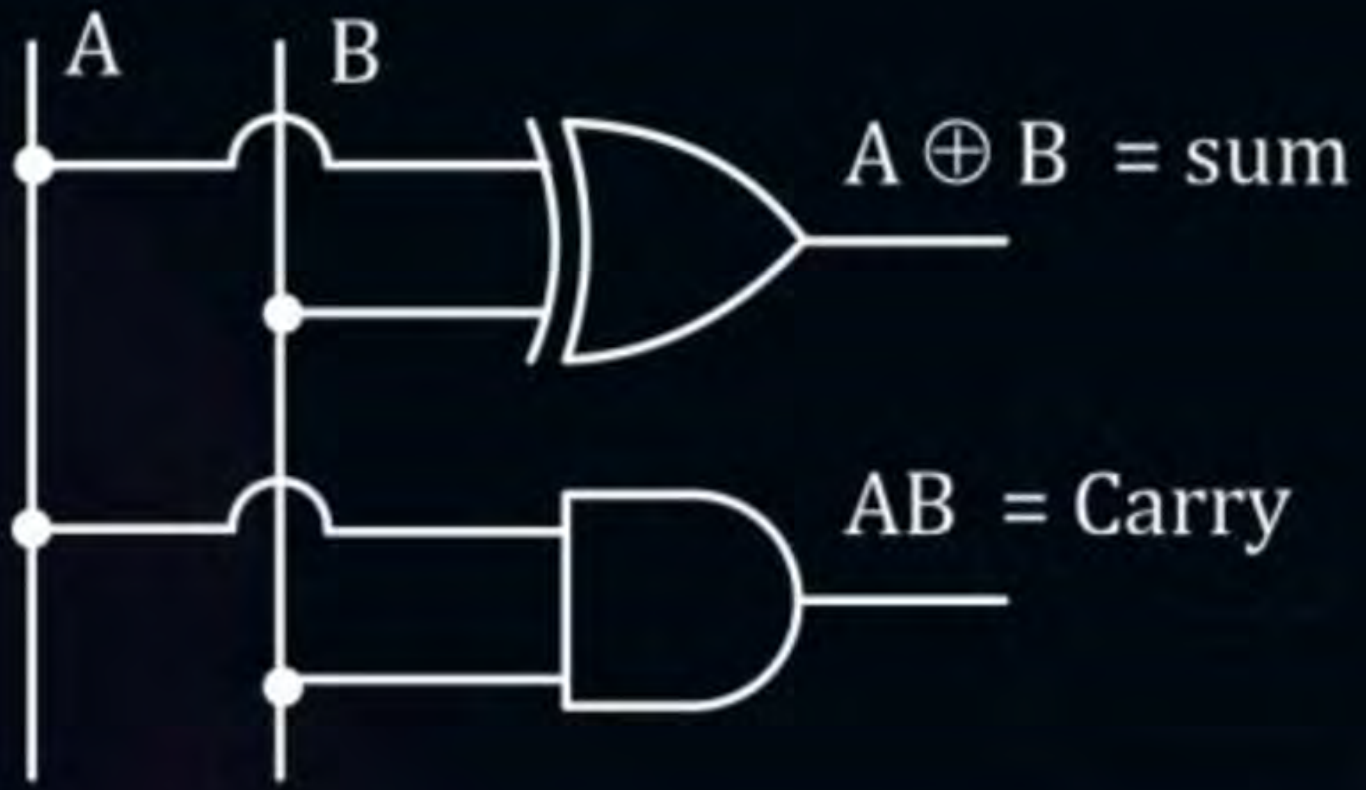
Minimization

No, already done

Topic : Half Adder

Topic : Half Adder

Step-5



min NAND

for H.A ?

Misconception:-

Min no. of NAND for H.A $\Longrightarrow$ HA $\Rightarrow$ XOR $\longrightarrow$ 4 NAND

\+

AND $\longrightarrow$ 2 NAND

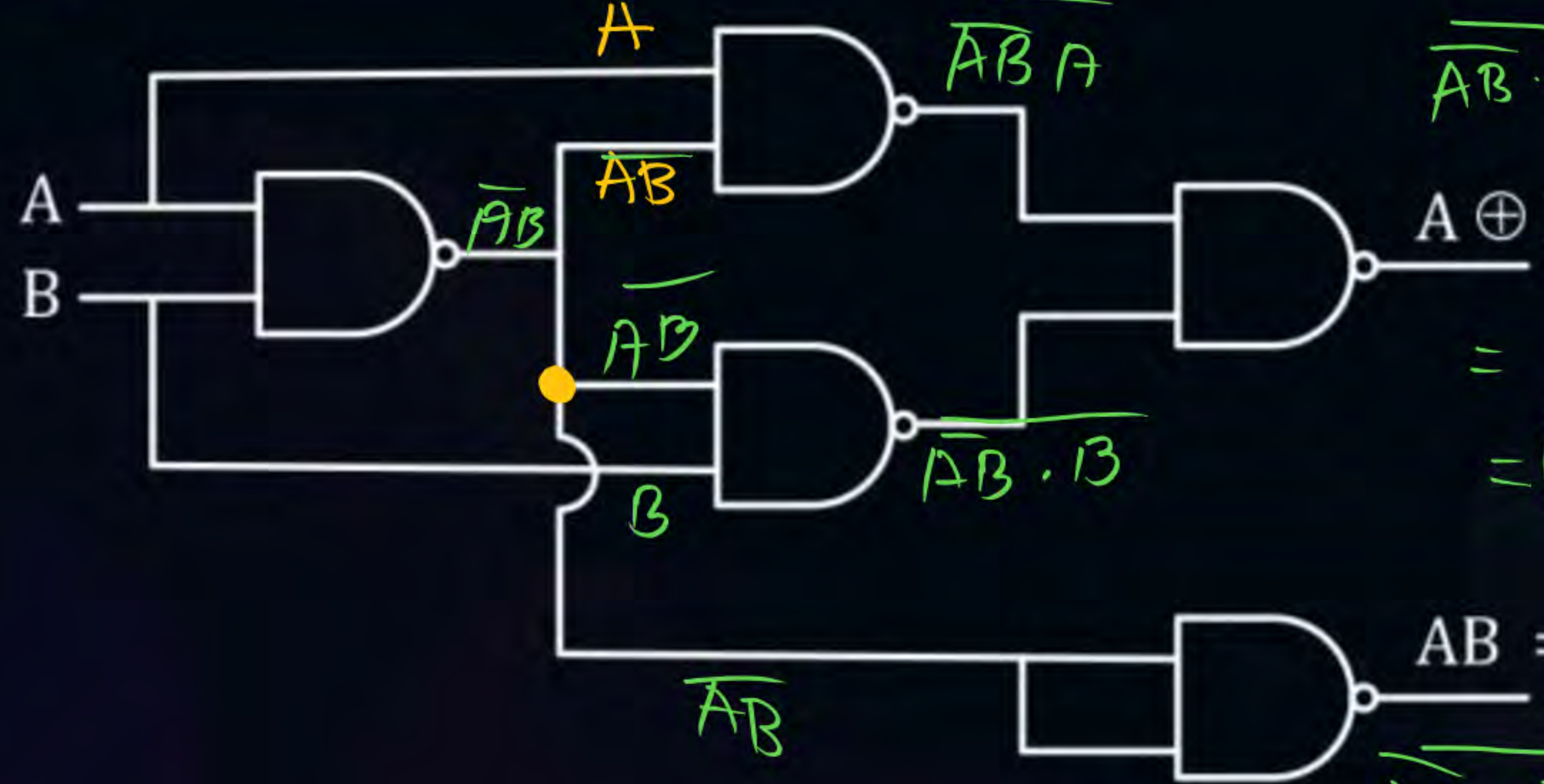
$\rightarrow 4+2 = 6$

(wrong)

Topic : Half Adder

HA

BY NAND GATE



$$\overline{\overline{\overline{AB}\cdot A}\cdot\overline{\overline{AB}\cdot B}}$$

$$= \overline{AB}\cdot A + \overline{AB}\cdot B$$

$$= (\bar{A}+\bar{B})A + (\bar{A}+\bar{B})B$$

$$= A \oplus B$$

$$\overline{\overline{AB}\cdot\overline{AB}} = AB$$

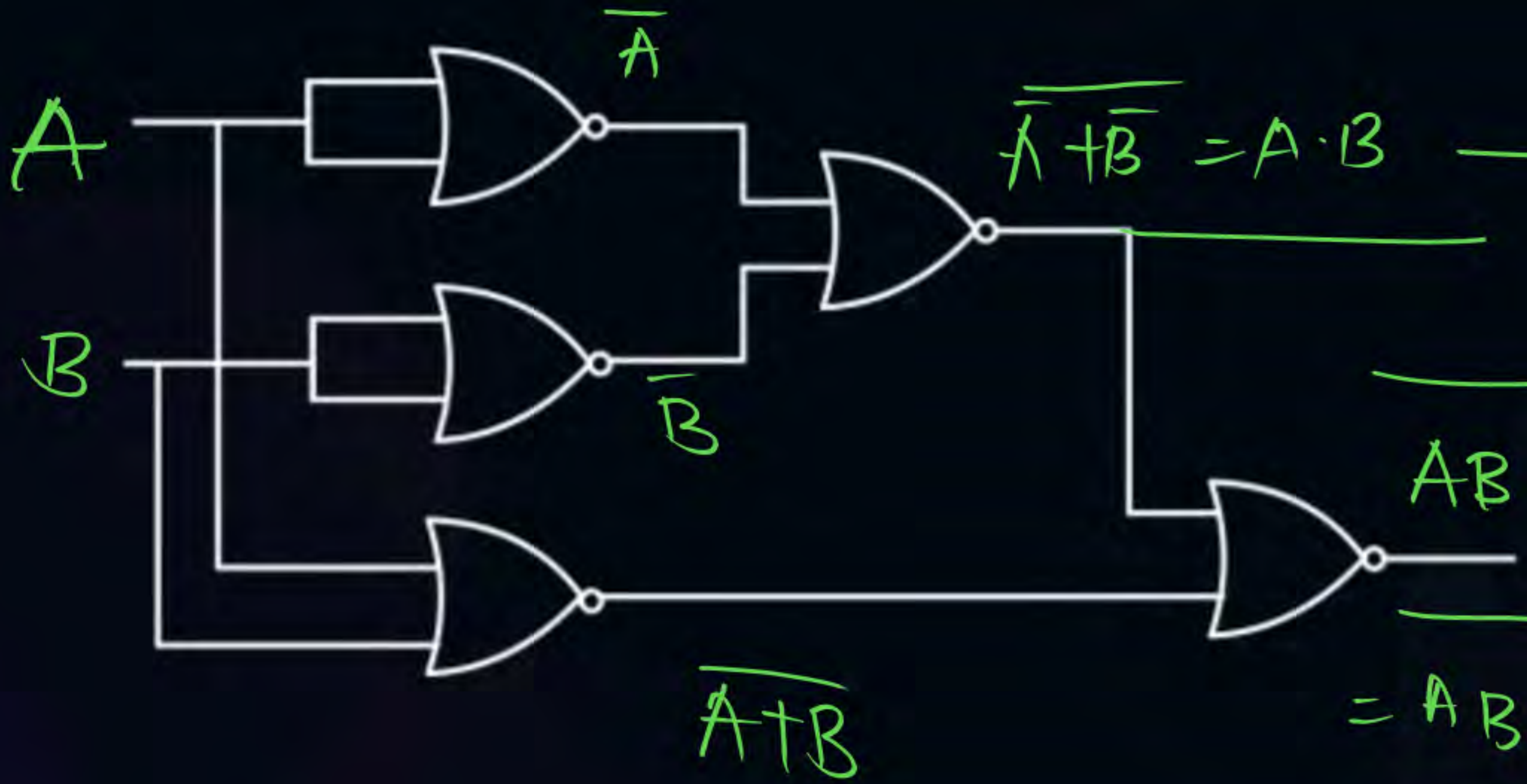
min. no. of NOR gates

for HA?

↳ Ans 5

Topic : Half Adder

By NOR GATE



$\overline{\bar{A}+\bar{B}} = A\cdot B \longrightarrow$ Carry

$\overline{AB + (\overline{A+B})}$

$= \overline{AB + \bar{A}\cdot\bar{B}}$

$= \overline{A \odot B} = A \oplus B \Rightarrow$ Sum

$\underline{\underline{5 \text{ NOR}}}$

HA

Shortcut :)

Anything HALF → 5

HA HS

NAN NOR NAND NOR

5

Anything FULL → 9

FA FS

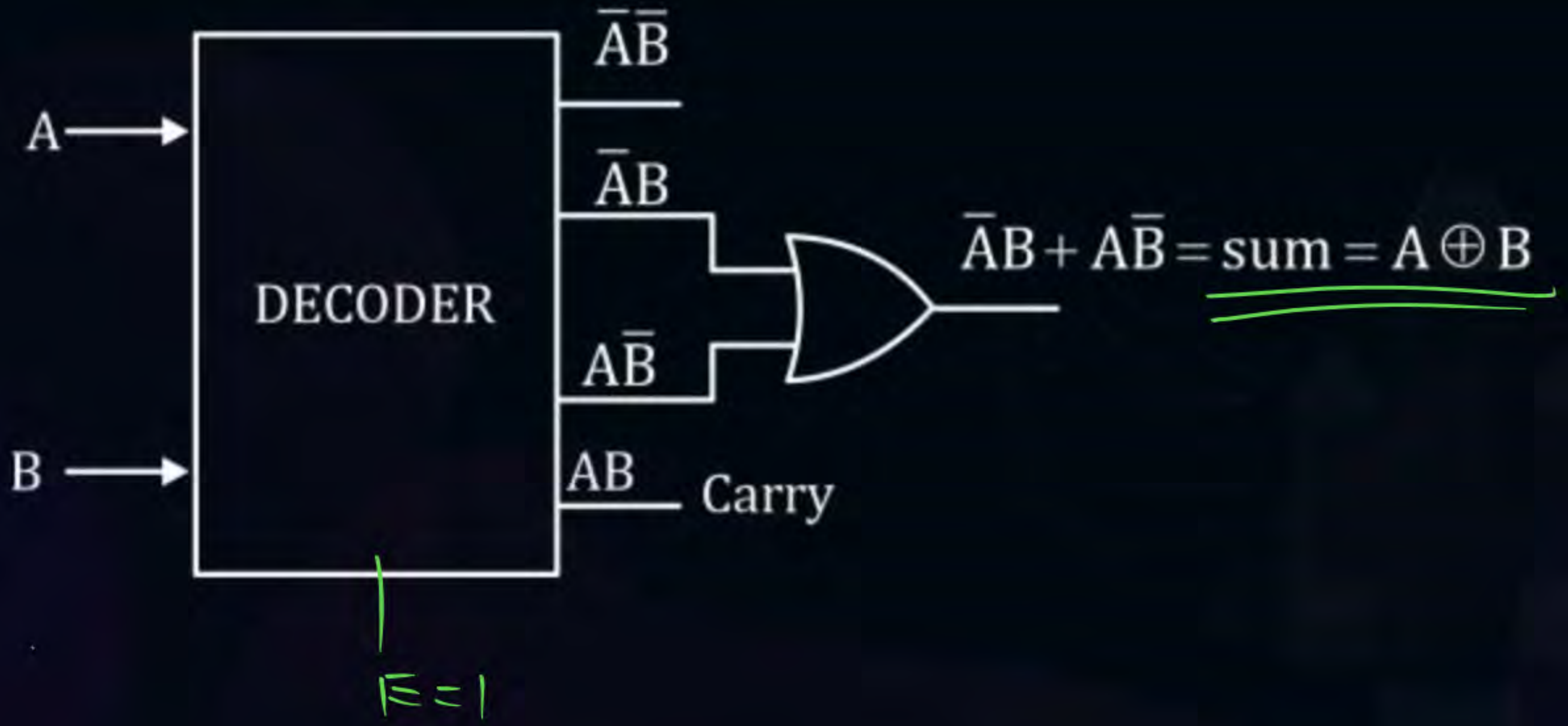
NAND NOR NAND NOR

9

(Q) Implement H.A using a 2X4 Decoder

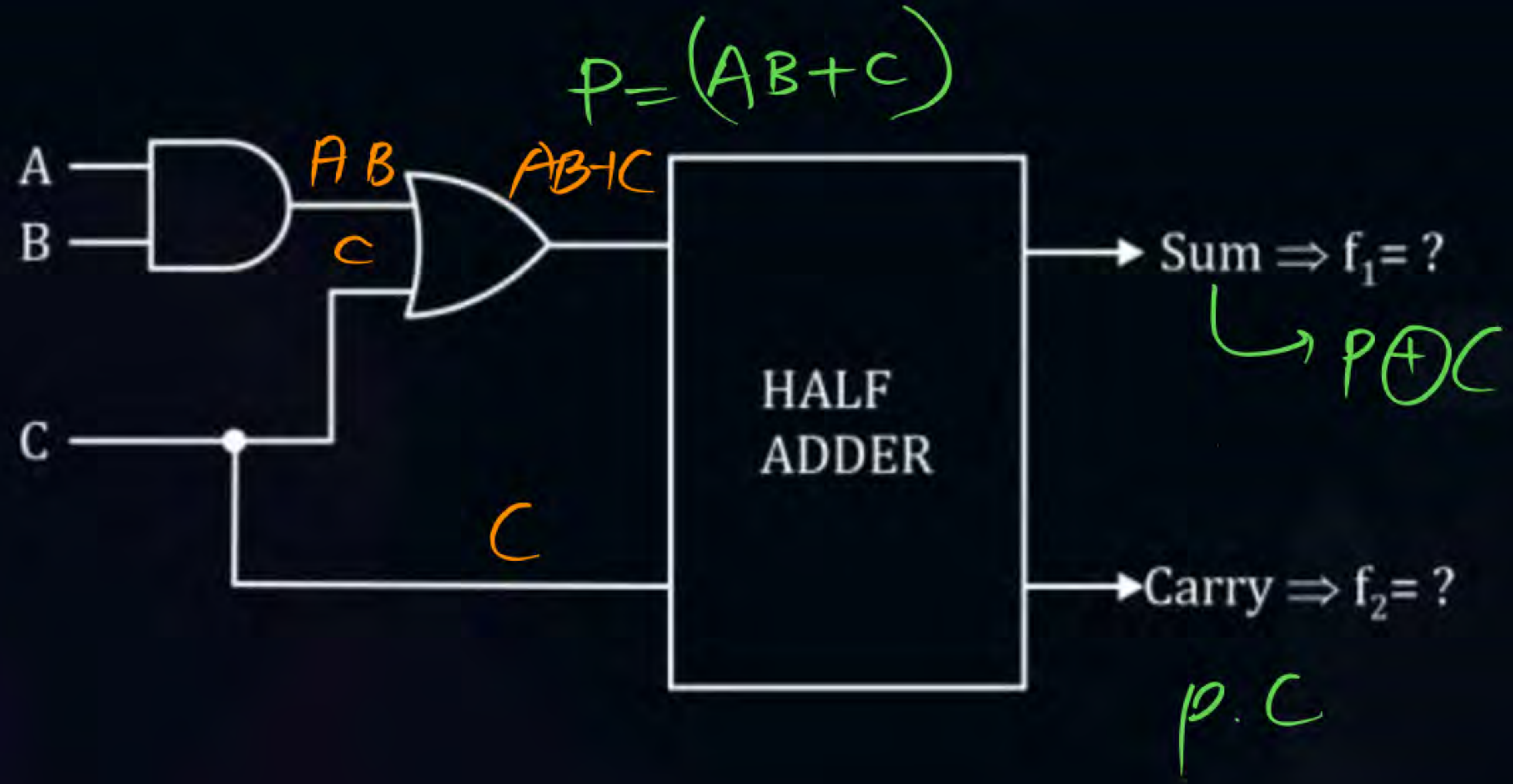
Topic : Half Adder

Half adder by using Decoder 2x4



[NAT]

#Q. What is the value of f_1 and f_2 ?



Ans
on Telegram

$$f_1 = \text{Sum} = P \oplus C$$

$$= \overline{P}C + P\overline{C}$$

$P = (AB + C)$

$$= \left(\overline{AB + C}\right)C + (AB + C)\overline{C}$$

$$= \cancel{\left(\overline{AB} \cdot \overline{C}\right)C} + AB\overline{C} + \cancel{C \cdot \overline{C}} = \underline{\underline{AB\overline{C}}}$$

$$f_2 = \text{Carry} = P \cdot C$$

$$= (AB + C) \cdot C$$

$$= ABC + C \cdot C$$

$$= ABC + C = (AB + 1) \cdot C = 1 \cdot C = \boxed{C}$$

Full Adder

(FA)

Topic : Full Adder

Three ~~bill~~ bits adder are known as full adder.

0	0	0	1
0	0	1	1
+0	+1	+1	+1
00	01	10	11

Carry Sum

$$\begin{array}{r} 0 \\ 1 \\ +\ 0 \\ \hline 01 \\ \hline \end{array}$$

Topic : Full Adder

Step-1

I/p → 3

O/p → 2

Topic : Full Adder

Step-2

	A	B	C	Sum	Carry
0	0	0	0	0	0
1	0	0	1	1	0
2	0	1	0	1	0
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	1
6	1	1	0	0	1
7	1	1	~~0~~ 1	1	1

V. Imp **Obs**: Carry = 1, when 2 or more i/ps are 1 out of 3 i/ps.

$\left. 2/3 \text{ or } 3/3 \right\}$

→ **Majority High input logic**

A B C

0 1 0 → 0

1 0 1 → 1

0 1 0

Topic : Full Adder

Step-3 Sum (A,B,C) $= \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$

$= \Sigma m(1,2,4,7)$

$= A \oplus B \oplus C$

Step-4 Carry (A,B,C) $= \Sigma m(3,5,6,7) = \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$

$= AB + BC + AC \rightarrow$ Minimize expression

$= (\bar{A}B + A\bar{B})C + AB(\bar{C} + C)$

$= \boxed{(A \oplus B)\, C + AB} \rightarrow$ Semi minimized

Imp

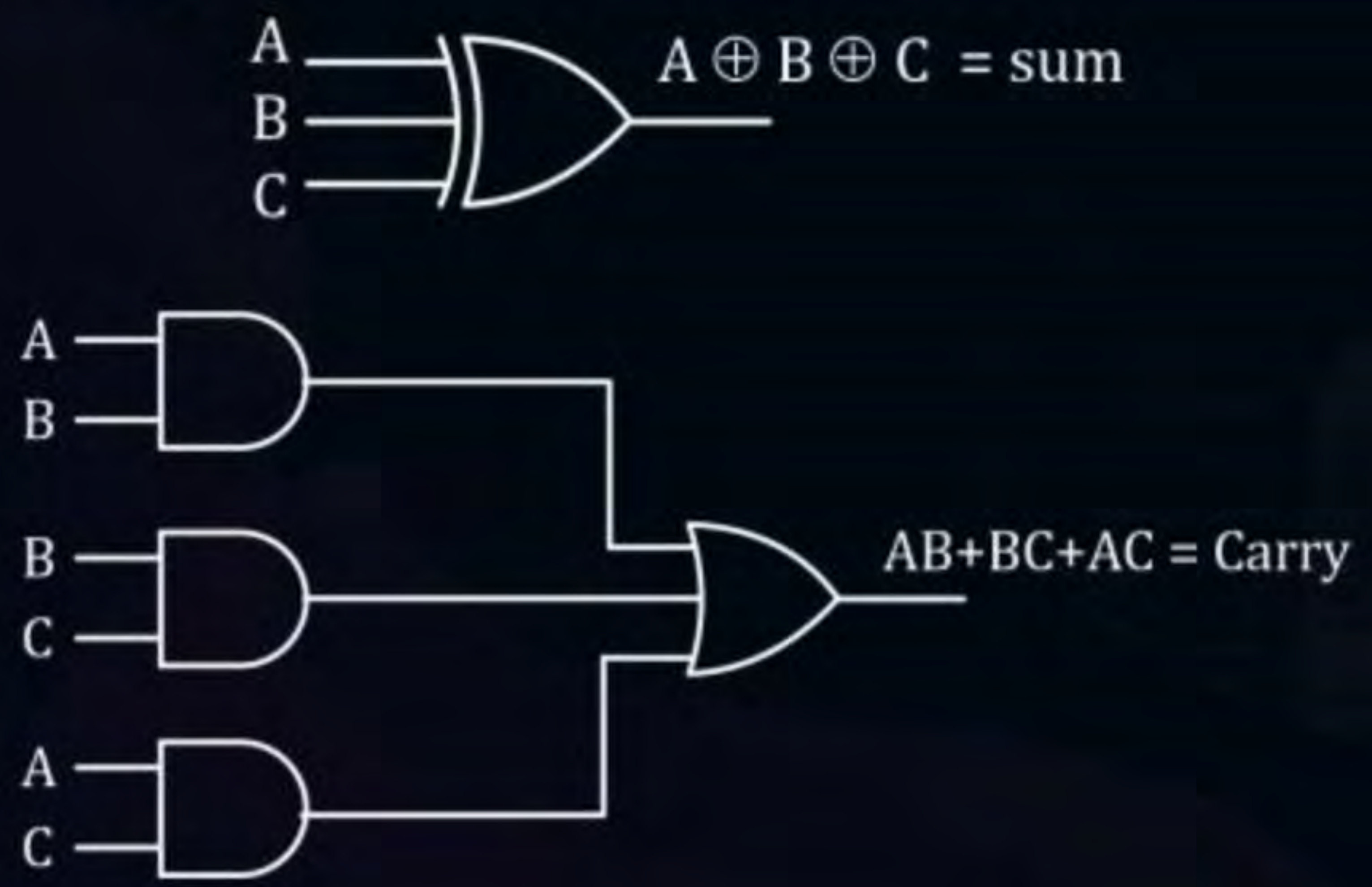
	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0	1	1 3	2
A	4	1 5	1 7	1 6

Topic : Full Adder

Step-5

$$\text{Sum} = A \oplus B \oplus C$$

$$\text{Carry} = AB + BC + AC$$



Imp obs:

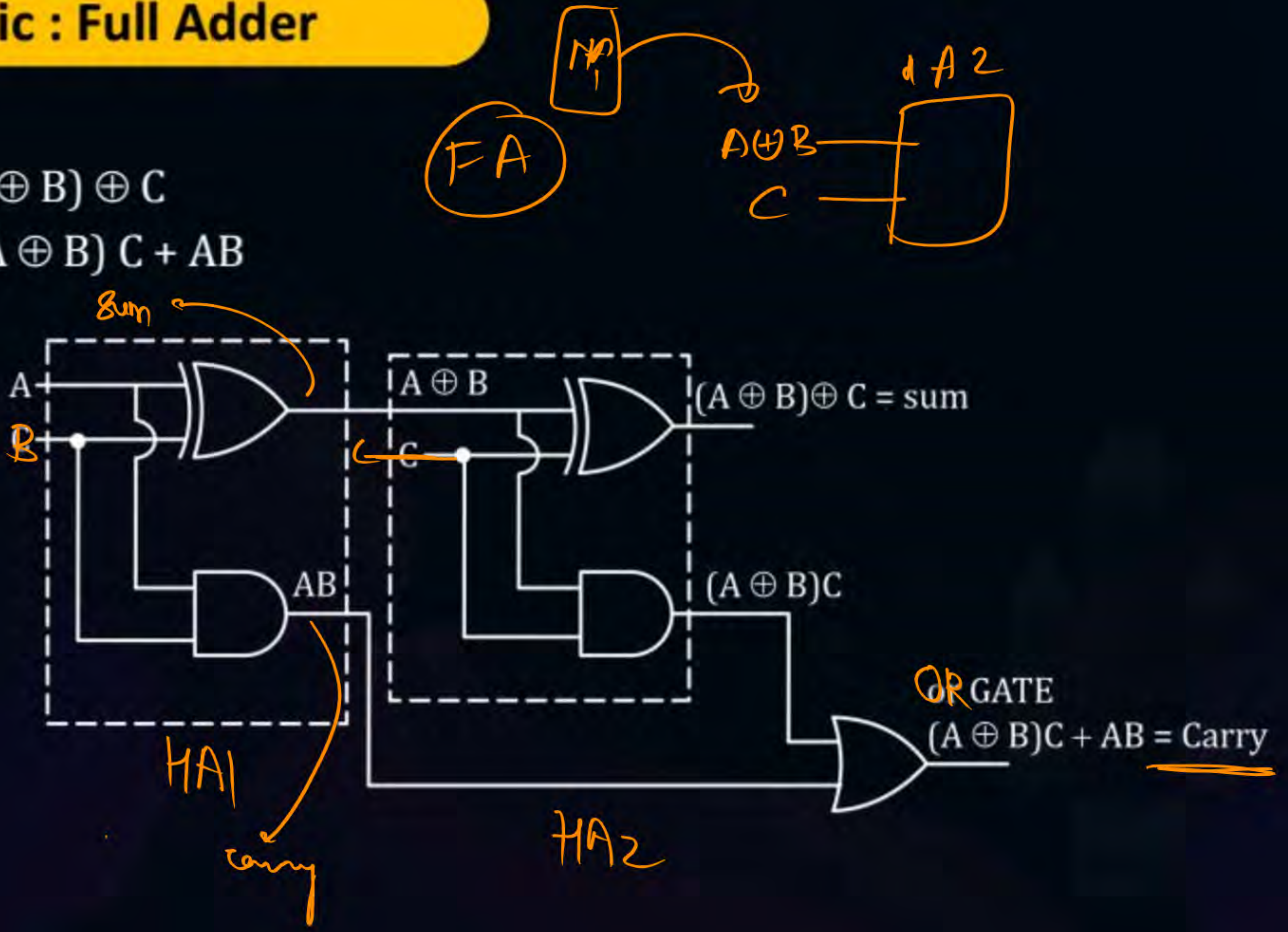
$$\text{Sum} = A \oplus B \oplus C$$

$$\text{Carry} = (A \oplus B) \cdot C + AB \longrightarrow \text{Semi-minimized}$$

Topic : Full Adder

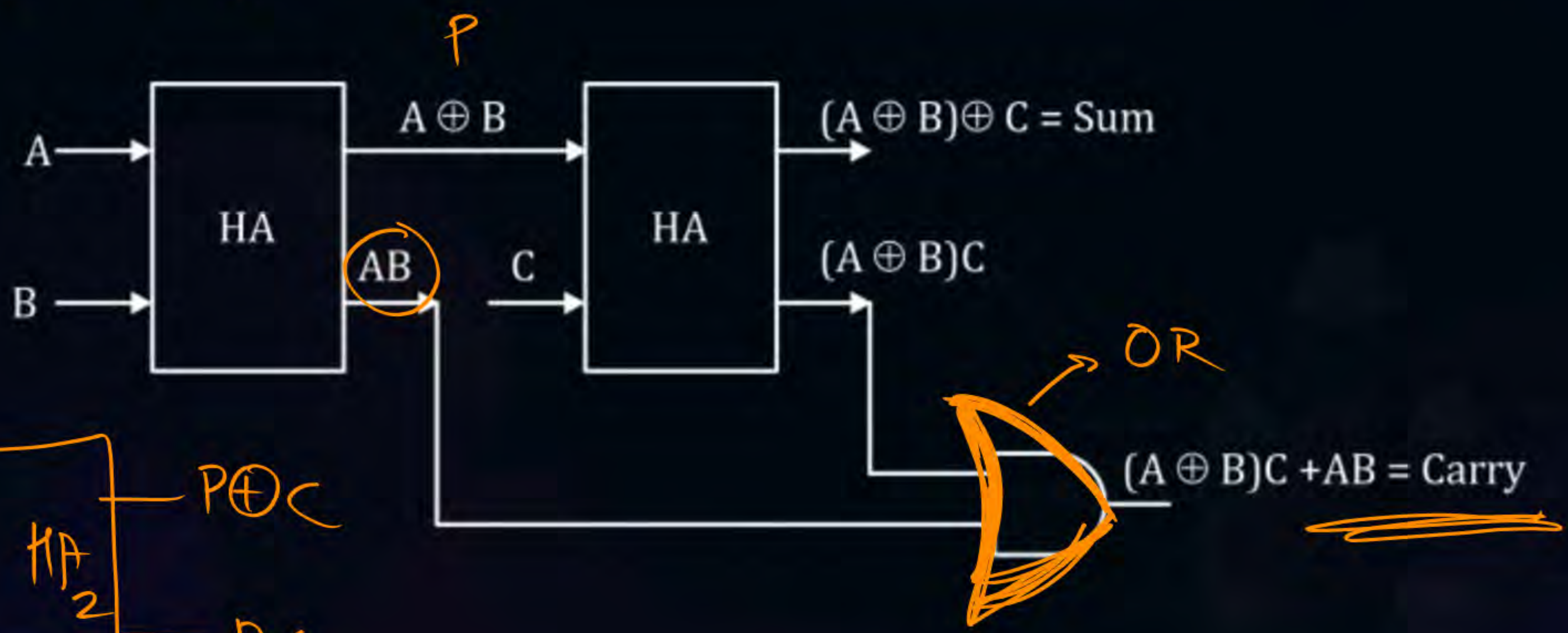
$$\text{Sum} = (A \oplus B) \oplus C$$

$$\text{Carry} = (A \oplus B)\,C + AB$$



Topic : Full Adder

Simplified Repr ⇒ FA using HA



Full Adder = 2 Half Adder

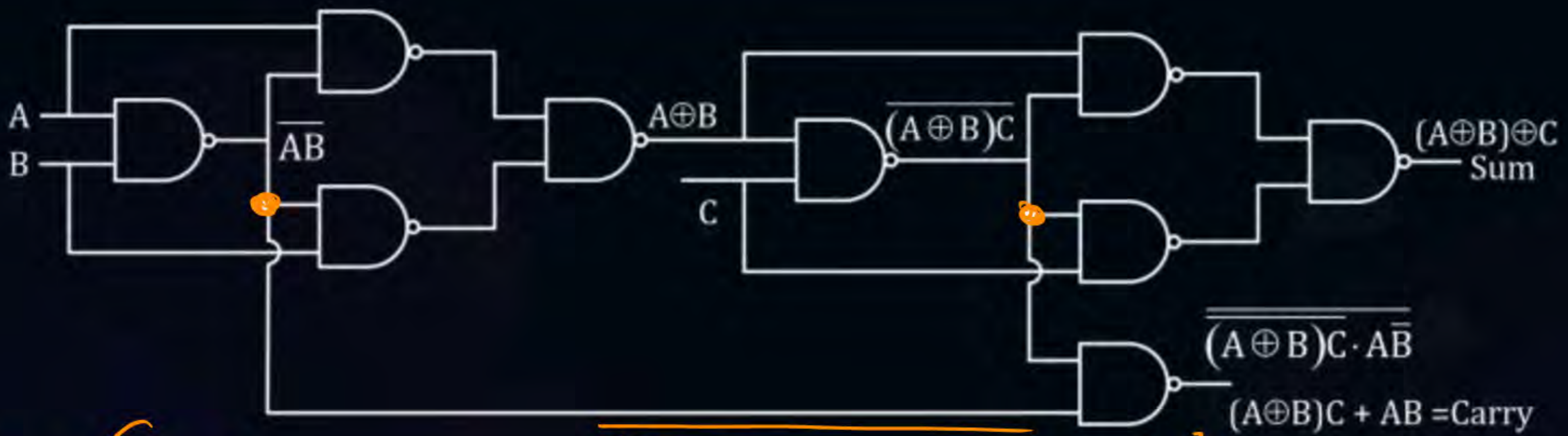
\+

1 OR Gate

Topic : Full Adder

Full adder by using NAND GATE

9 NAND



$\text{Carry} = (A\oplus B)\cdot C + AB$

$\overline{\overline{(A\oplus B)\cdot C}\cdot\overline{AB}}$

[NAT]

#Q. Full order by using NOR GATE

9

H.W

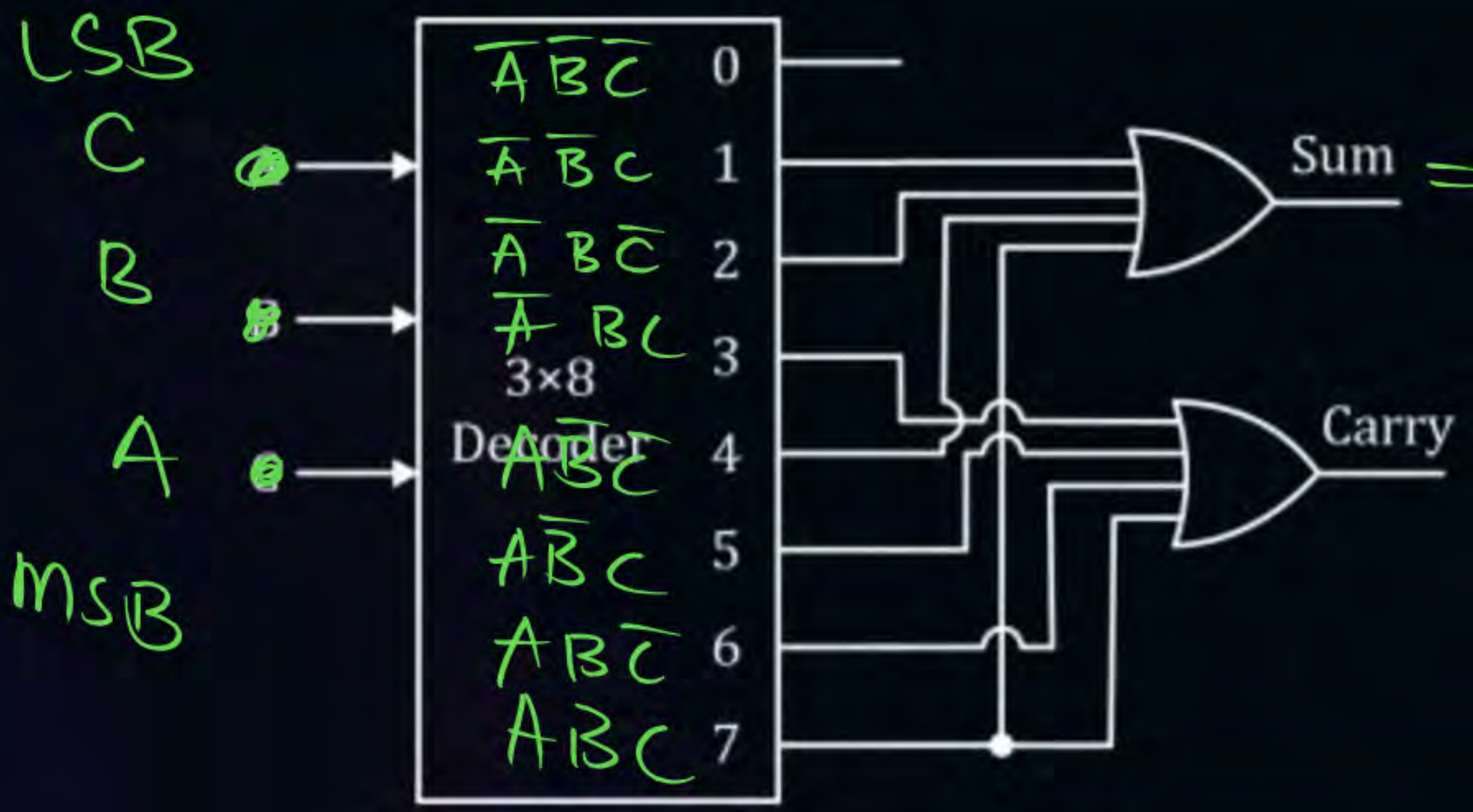
(Q)

Represent FA using a 3X8 decoder

Topic : Full Adder

Full adder by using decoder

3×8

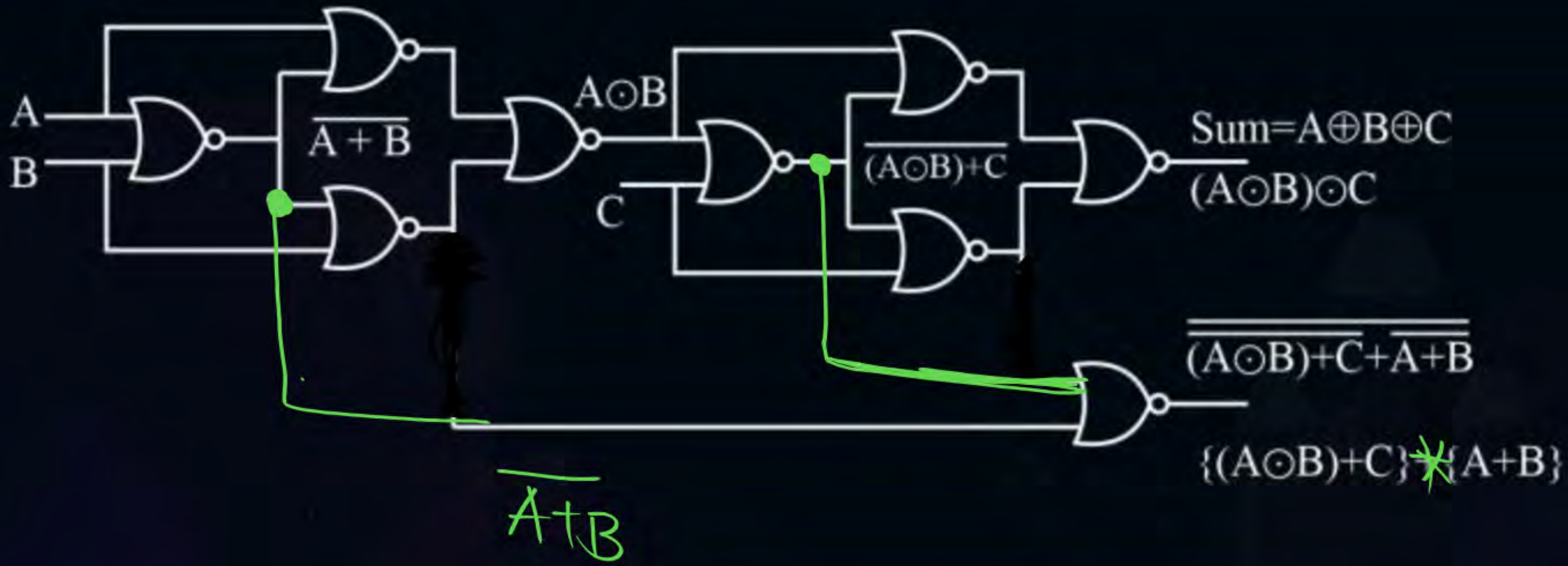


Sum $= A \oplus B \oplus C$

Carry $= AB + BC + AC$

Topic : Full Adder

H.W Ans Full Adder using NOR



FA using NOR

as per prev image,

$$\text{Sum} = A \odot B \odot C = A \oplus B \oplus C \checkmark$$

$$\text{Carry} = \left((A \odot B) + C\right)(A + B)$$

$$= \left((AB + \bar{A}\bar{B}) + C\right)(A + B)$$

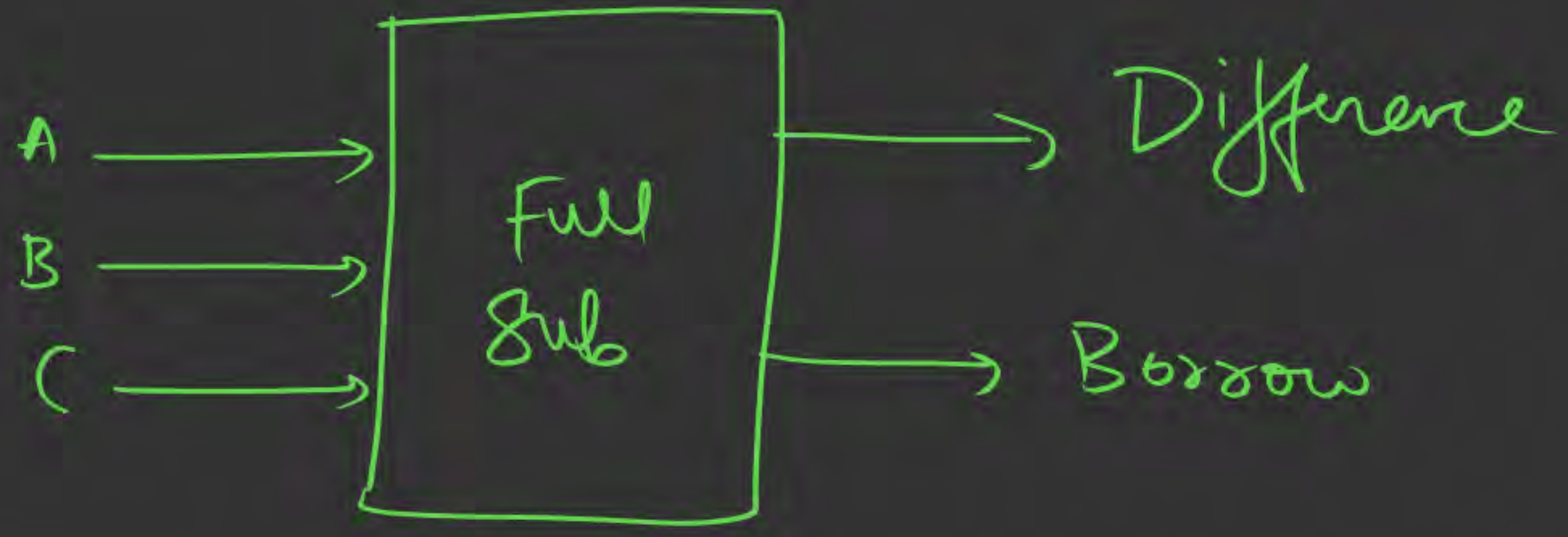
$$= ABA + ABB + \cancel{\bar{A}\bar{B}A} + \cancel{\bar{A}\bar{B}B} + AC + BC$$

$$= AB + AB + AC + BC$$

$$= AB + BC + AC$$

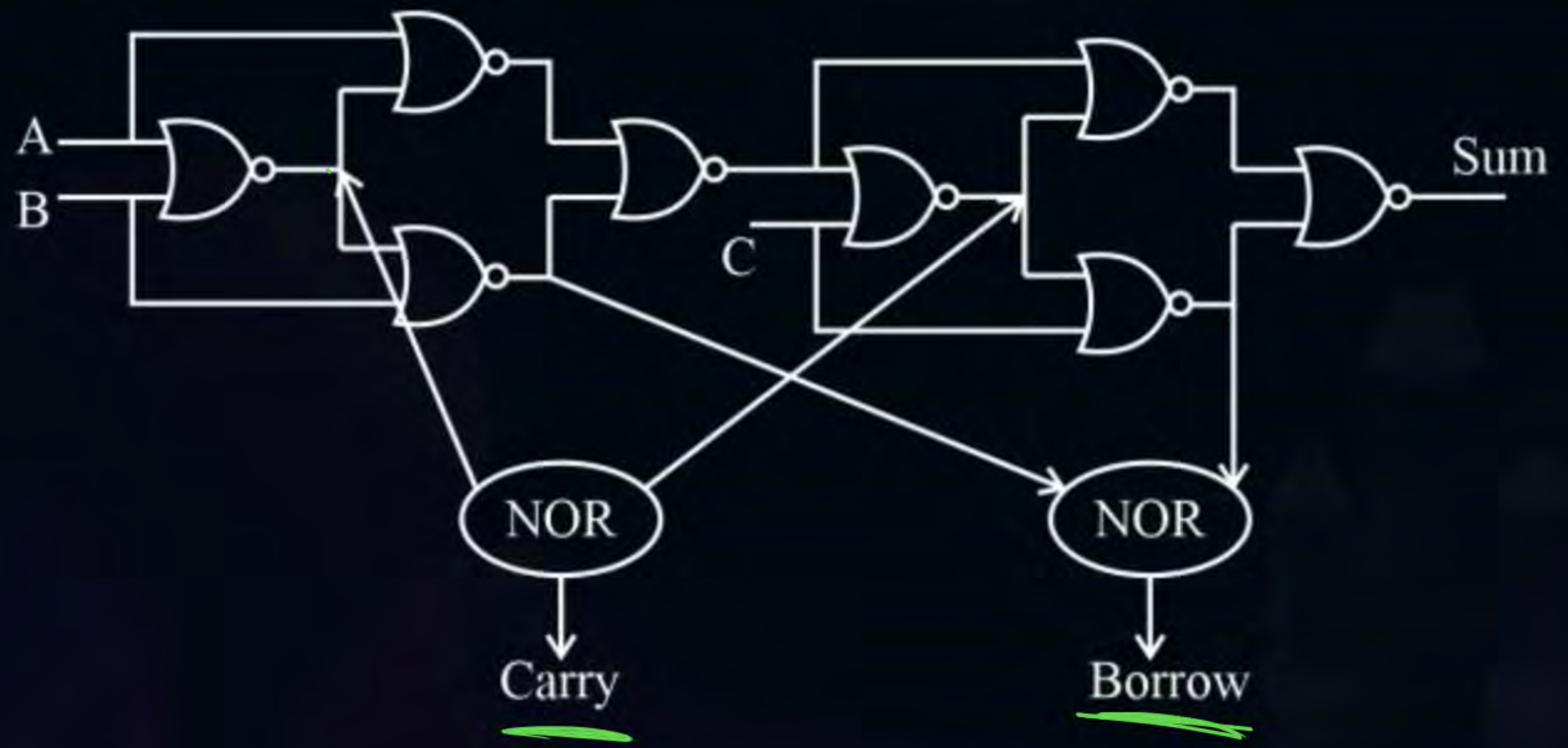
Full Subtractor (FS)

↳ minor changes in the NOR impl of FA.


A
B
C
Full
Sub
Difference
Borrow

Topic : Full Adder

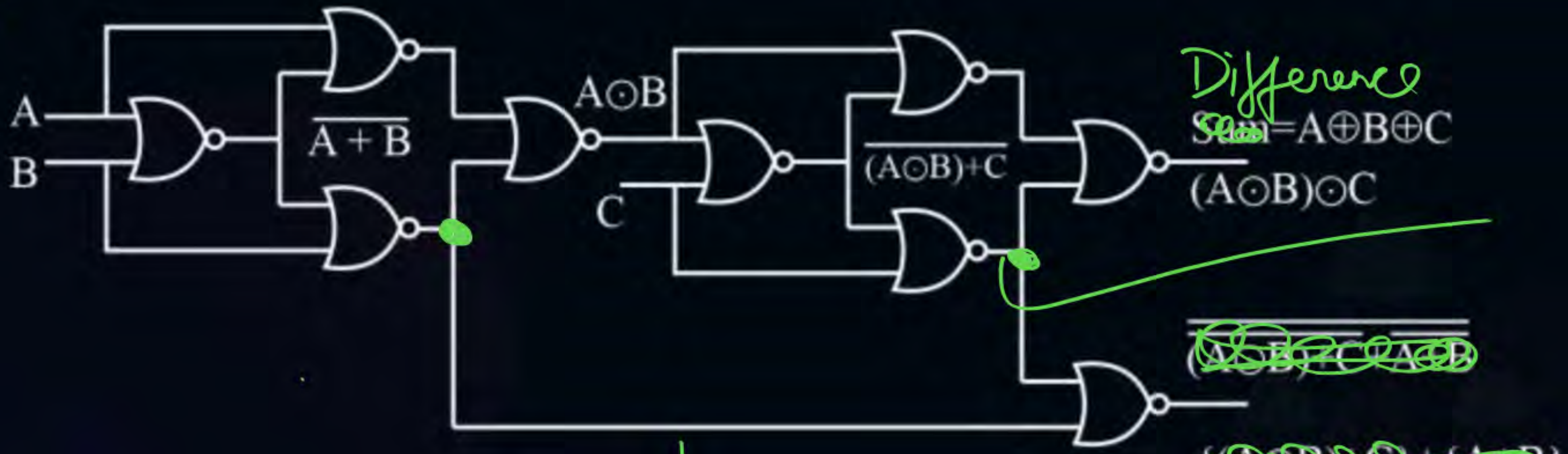
(FA & FS using NOR)



Topic : Full Adder

H.W Ans — Full Subtractor using NOR

(Bonus)



$$\overline{\overline{A+B}+B} = \overline{\bar{A}\,\bar{B}+B} = \overline{\bar{A}\,\bar{B}} \cdot \bar{B} = (A+B)\cdot\bar{B}$$

$$= A\bar{B}$$

Topic : Full Adder

$$\overline{A\bar{B} + (A \odot B)\bar{C}}$$

$$\overline{A\bar{B}} \cdot \overline{(A \odot B)\bar{C}}$$

$$(\bar{A} + B)\left[\overline{(A \odot B)} + C\right]$$

$$(\bar{A} + B)[\bar{A}B + A\bar{B} + C]$$

$$\bar{A}B + \bar{A}C + \bar{A}B + BC$$

$$\boxed{\bar{A}B + \bar{A}C + BC \Rightarrow \boxed{\text{Brrow}}}$$

Shortcut

F·A carry $\Rightarrow AB + BC + AC$

F·S borrow $\Rightarrow \bar{A}B + BC + \bar{A}C$

Topic : Full Adder

$\boxed{\text{Summary for } \underline{\underline{\text{GATE}}}}$

Half adder

Sum = $A \oplus B$

Carry = AB

NAND /NOR = 5

Full adder

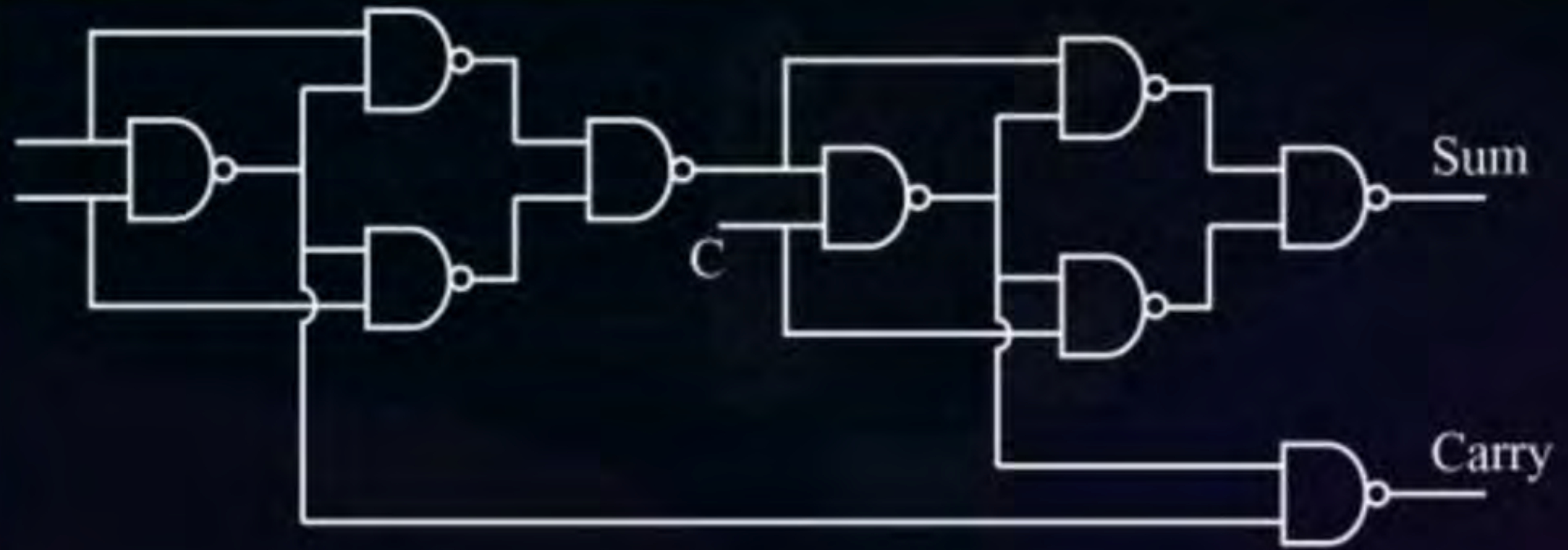
NAND/NOR = 9

Sum = $\sum m\{1,2,4,7\} = A \oplus B \oplus C$

Carry = $\sum m\,(3,5,6,7)$

$= (A \oplus B)\,C + AB \rightarrow$ semi- minimized

$= \underline{\underline{AB + BC + AC}}$



Topic : Half Subtractor

Two bit subtractor are known as half subtractor.

Step-1

2 i/p
2 o/p

Topic : Half Subtractor

Step-1

A	B	Diff.	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Step-2

$$\begin{array}{r} 0 \\ -\ 0 \\ \hline 0\ 0 \end{array} \qquad \begin{array}{r} 1 \\ -\ 0 \\ \hline 0\ 1 \end{array} \qquad \begin{array}{r} 1 \\ -\ 1 \\ \hline 0\ 0 \end{array}$$

$$\star \quad \begin{array}{r} 0 \\ -\ 1 \\ \hline 1\ 1 \end{array}$$

Borrow Diff

Topic : Half Subtractor

Step-3

Diff. $\underline{\underline{A \oplus B}}$

$$\begin{array}{r} 0 \\ -\ 1 \\ \hline 1\ 1 \\ \hline \end{array}$$

Borrow: $\bar{A}B$

Shortcut :-)

H.A → Carry $A \cdot B$

H.S → Borrow: $\bar{A}B$

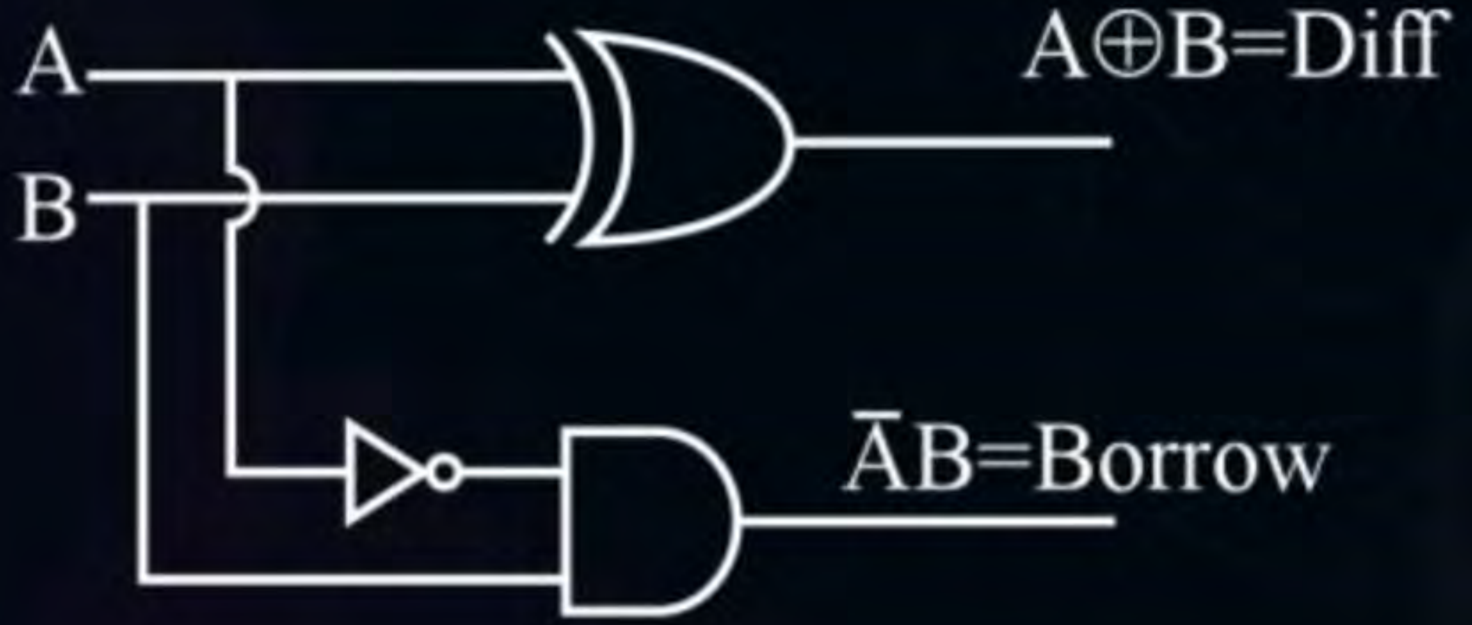
F.A
H.A } Replace $A \Rightarrow \bar{A}$, you get Borrow

Topic : Half Subtractor

Step-4

No minimization

Step-5



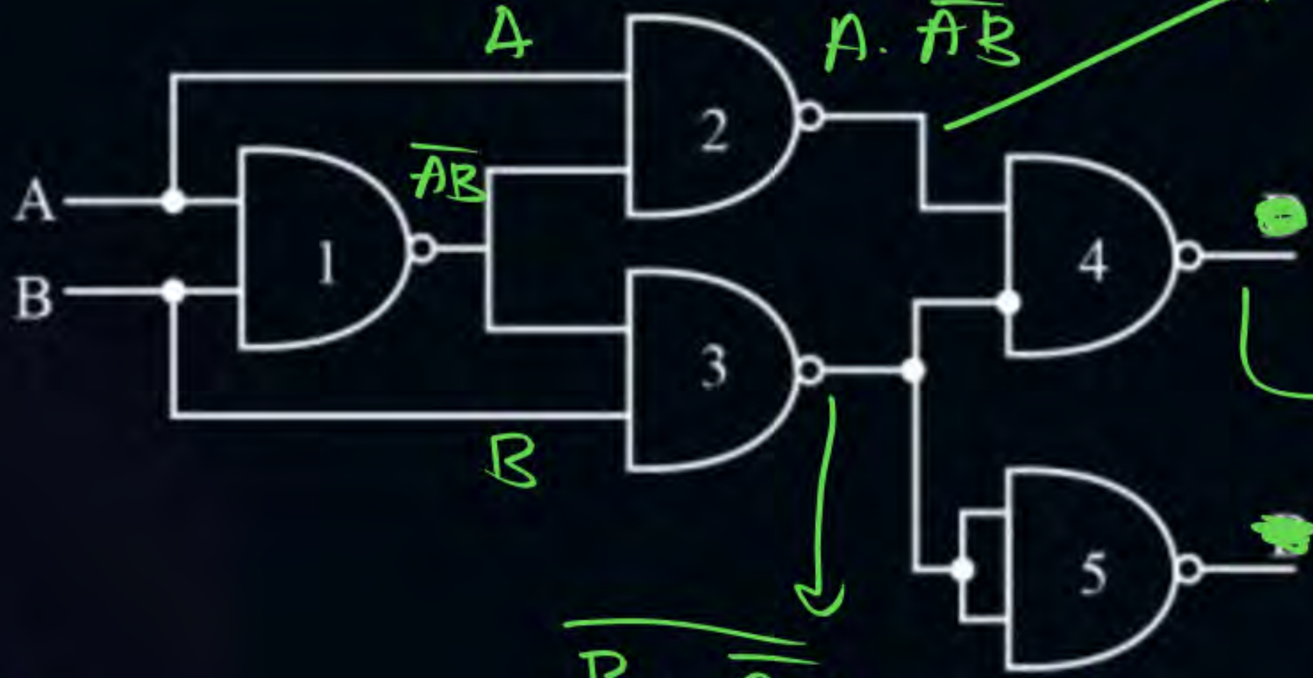
Topic : Half Subtractor

$\overline{A+\overline{B}} = \overline{A}\cdot B$

5 NAND

$\overline{A\cdot A} = \overline{A}$

(Not gate)



$\overline{A} + \overline{\overline{AB}} = \overline{A} + AB$

$= (\overline{A}+A)(\overline{A}+B)$

$= \overline{A}+B$

$\overline{(\overline{A}+B)\cdot(A+\overline{B})}$

$= \overline{\overline{A}+B} + \overline{A+\overline{B}}$

$= A\cdot\overline{B} + \overline{A}\cdot B$

$= A\oplus B \rightarrow$ Diff

$\overline{B\cdot\overline{AB}}$

$= \overline{B} + \overline{\overline{AB}}$

$= \overline{B} + AB$

$= (\overline{B}+A)(\overline{B}+B) = A+\overline{B}$

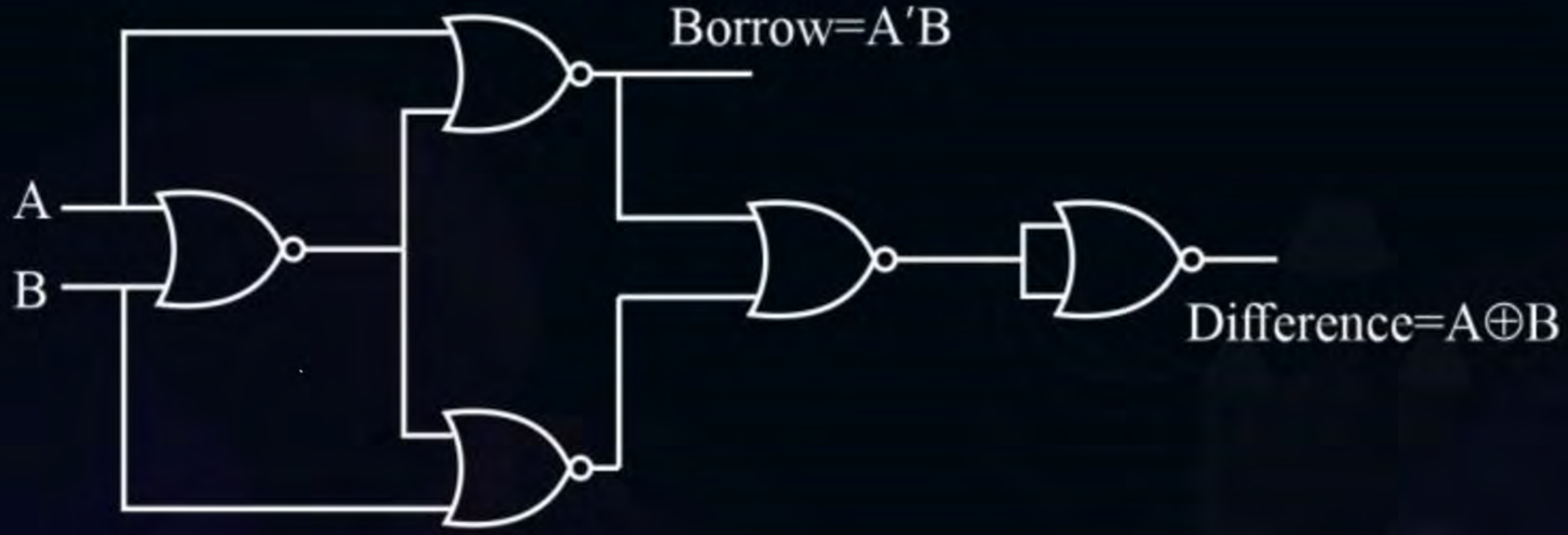
$\overline{A+\overline{B}} = \overline{A}B \rightarrow$ Borrow

Half Subtractor using NAND gate

Topic : Half Subtractor

H.S using NOR

H.W



Topic : Full Subtractor

Topic : Full Subtractor

H.W

INPUT			OUTPUT	
A	B	C	D	Borrow
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Topic : Full Subtractor

H.W

HA

$\text{Sum} = A \oplus B$

$\text{Carry} = AB$

HS

$\text{Diff. } A \oplus B$

$\text{Borrow: } \bar{A}B$

$F \rightarrow \bar{A}$

FA

$\text{Sum} = A \oplus B \oplus C$

$\text{Carry} = AB + AC + BC$

HS

$\text{Diff. } A \oplus B \oplus C$

$\text{Borrow: } \bar{A}B + \bar{A}C + BC$

$A \rightarrow \bar{A}$

2 mins Summary

→ HA

FA

HS

FS → Revise Tomorrow

THANK - YOU